Frm(L0) is exactly the strings with a formation sequence

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-propositional-logic-syntax-and-semantics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof carries the conjunction case in full and closes the other four connective cases by explicitly parallel reasoning; the lane precedent (registered FOL twin X1-logic-fol-synsem-formed, and the simpc splitting-sets family) accepts one fully-carried representative case when the rest are notational variants.

Editorial note. The helper lem:fseq-init is cited by the proof but is an Exercise in the source and is not a counted card; the dependency is recorded as a prerequisite, not silently dropped.

Editorial note. The label thm:fseq-frm-equiv names the counted statement; the counted statement is carried by the author's own thm environment. A same-label theorem in the first-order-logic-syntax-and-semantics unit is a distinct unit-scoped statement (7 connective cases) and is a separate registered card (L46216); cross-unit same-name labels are chapter-scoped per the fitting-ideal precedent.

Allowed standard prerequisites (editorial).

- Every formula has a formation sequence. Same unit, \ollabel{prop:formed} (source line 589, own card). Line 648-649: gives the easy inclusion Frm(L0) subseteq F before the converse.
- Initial segments of a formation sequence are formation sequences. Same unit, \ollabel{lem:fseq-init} (source line 629; Exercise in source). Line 667-669: truncates the sequence at j and k for the induction hypothesis.

Source reference. Original proposition formed: sources/olp-propositional-logic-syntax-and-semantics/pinned-original-source.tex, source lines 589–589. Prerequisite pointer invoked inside the counted proof; it names the formula-formation proposition of the same unit whose content is standard and fully retained in the unit, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma fseq-init: sources/olp-propositional-logic-syntax-and-semantics/pinned-original-source.tex, source lines 629–629. Prerequisite pointer invoked inside the counted proof; it names the initial-formula-sequence lemma of the same unit, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. $\mathrm{Frm}(\mathcal{L}_0)$ *is the set of all strings of symbols in the language* $\mathcal{L}L_0$ *with a formation sequence.*

Proof. Let F be the set of all strings of symbols in the language $\mathcal{L}L_0$ that have a formation sequence. We have seen in formed that $\mathrm{Frm}(\mathcal{L}_0) \subseteq F$, so now we prove the converse.

Suppose A has a formation sequence $\langle A_0, \ldots, A_n \rangle$. We prove that $A \in \mathrm{Frm}(\mathcal{L}_0)$ by strong induction on n. Our induction hypothesis is that every string of symbols with a formation sequence of length $m < n$ is in $\mathrm{Frm}(\mathcal{L}_0)$. By the definition of a formation sequence, either A_n is atomic or there must exist $j, k < n$ such that one of the following is the case:

1. $A_n \equiv \neg A_j$.
2. $A_n \equiv (A_j \wedge A_k)$.
3. $A_n \equiv (A_j \vee A_k)$.
4. $A_n \equiv (A_j \to A_k)$.
5. $A_n \equiv (A_j \leftrightarrow A_k)$.

Now we reason by cases. If A_n is atomic then $A_n \in \mathrm{Frm}(\mathcal{L}_0)$. Suppose instead that $A \equiv (A_j \wedge A_k)$. By fseq-init, $\langle A_0, \ldots, A_j \rangle$ and $\langle A_0, \ldots, A_k \rangle$ are formation sequences for A_j and A_k respectively. Since these are proper initial subsequences of the formation sequence for A, they both have length less than n. Therefore by the induction hypothesis, A_j and A_k are in $\mathrm{Frm}(\mathcal{L}_0)$, and so by the definition of formula, so is $(A_j \wedge A_k)$. The other cases follow by parallel reasoning. $\square$